Data Source Evaluation Report

How ready your sources were, and what was left unmapped

Date: 2026-11-09

Project: test-claude-plugin-35 (Customer 360, synthetic)

Summary

- Total data sources: 3
- Entity Specification-compliant: 0
- Needs mapping: 3
- Needs enrichment: 0

No source carries a record type, a date of birth or an identifier; matching will rest on name, address, phone and email.

Data Source Details

Data Source 1: CRM_SYSTEM

Status: Needs mapping

Location: data/raw/crm_system.csv

Records: 4,000

Fields: 7 columns

Quality: 94.8% (completeness 94.7%, format consistency 94.0%, duplicate rate 0.0%) - band >=80

Evaluation:

- crm_id is the record key (RECORD_ID); DATA_SOURCE is derived as CRM_SYSTEM.
- full_name is a single name field, so it corresponds to NAME_FULL (the specification says not to parse it).
- address is a single free-text field, so it corresponds to ADDR_FULL.
- phone corresponds to PHONE_NUMBER; email to EMAIL_ADDRESS.
- created_date and campaign have no specification counterpart (payload candidates).
- 300 records are campaign duplicates of another CRM record, each with its own key.

Reason: Field names differ from the Entity Specification and the records are flat CSV rows, not Senzing JSON.

Record type check: 0 of 0 PERSON-typed records are type/name candidates (no record-type field). Decision: none needed

Next step: Phase 2 (mapping)

Data Source 2: BILLING_SYSTEM

Status: Needs mapping

Location: data/raw/billing_system-regenerated.csv (original: data/raw/billing_system.csv)

Records: 4,000

Fields: 10 columns

Quality: 90.2% (completeness 86.0%, format consistency 100.0%, duplicate rate 0.0%) - band >=80. History: 76.0 as collected; 77.7 after Step 7a format normalization; 90.2 after regeneration on the return to Data collection.

Evaluation:

- billing_id is the record key; DATA_SOURCE is derived as BILLING_SYSTEM.
- first_name / last_name are parsed name parts (NAME_FIRST / NAME_LAST).
- street, city, state, zip are parsed address parts (ADDR_LINE1, ADDR_CITY, ADDR_STATE, ADDR_POSTAL_CODE).
- phone corresponds to PHONE_NUMBER, in four formats.
- plan and last_invoice_date have no specification counterpart (payload candidates).
- As collected, the address block was absent on about 45% of records and phone on about 61%. The regenerated file narrows that to about 18% and 36%.

Reason: Field names differ from the Entity Specification and the records are flat CSV rows, not Senzing JSON.

Record type check: 0 of 0 PERSON-typed records are type/name candidates (no record-type field). Decision: none needed

Next step: Phase 2 (mapping)

Data Source 3: SUPPORT_TICKETS

Status: Needs mapping

Location: data/raw/support_tickets.csv

Records: 2,000

Fields: 5 columns

Quality: 95.7% (completeness 96.3%, format consistency 93.0%, duplicate rate 0.0%) - band >=80

Evaluation:

- ticket_id is the record key; DATA_SOURCE is derived as SUPPORT_TICKETS.
- name is a single name field (NAME_FULL), with inconsistent casing and occasional typos.
- email corresponds to EMAIL_ADDRESS, with occasional typos.
- subject and opened_date have no specification counterpart (payload candidates).
- 400 records are repeat tickets from someone who already filed one.

Reason: Field names differ from the Entity Specification and the records are flat CSV rows, not Senzing JSON.

Record type check: 0 of 0 PERSON-typed records are type/name candidates (no record-type field). Decision: none needed

Next step: Phase 2 (mapping)

Mapping Priority

1. CRM_SYSTEM - strongest source with the widest feature set (name, address, phone, email); it links to Billing by phone and name and to Support by email. 2. SUPPORT_TICKETS - strong but sparse; its email links it to CRM. 3. BILLING_SYSTEM - regenerated to 90.2; parsed name and address parts, so the mapping is mostly direct.

Cross-Source Outlook

Distinct values compared after light normalization (phone: last 10 digits; email: lowercase; name: lowercase letters, single-letter initials dropped). Exact-value overlap is a floor: nicknames, typos and formatting variants that Senzing can still match are not counted. Names include 20 deliberate pairs of different people who share a name.

- CRM_SYSTEM × BILLING_SYSTEM - measured
  - Shared attribute: phone
  - Distinct values shared: 1,226 (of 3,299 in CRM_SYSTEM, 2,546 in BILLING_SYSTEM)
- CRM_SYSTEM × BILLING_SYSTEM - measured
  - Shared attribute: name
  - Distinct values shared: 2,020 (of 3,700 in CRM_SYSTEM, 3,713 in BILLING_SYSTEM)
- CRM_SYSTEM × SUPPORT_TICKETS - measured
  - Shared attribute: email
  - Distinct values shared: 728 (of 3,351 in CRM_SYSTEM, 1,518 in SUPPORT_TICKETS)
- CRM_SYSTEM × SUPPORT_TICKETS - measured
  - Shared attribute: name
  - Distinct values shared: 848 (of 3,700 in CRM_SYSTEM, 1,604 in SUPPORT_TICKETS)
- BILLING_SYSTEM × SUPPORT_TICKETS - measured
  - Shared attribute: name
  - Distinct values shared: 717 (of 3,713 in BILLING_SYSTEM, 1,604 in SUPPORT_TICKETS)